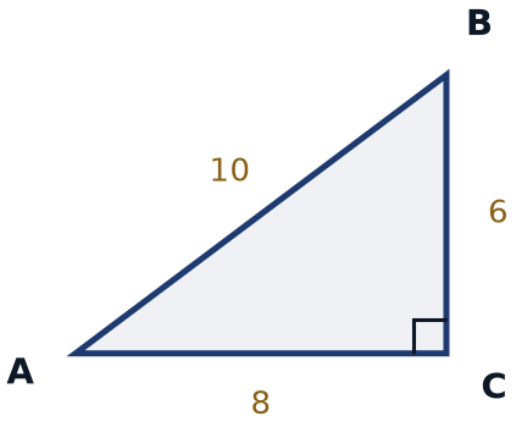
Calculus

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

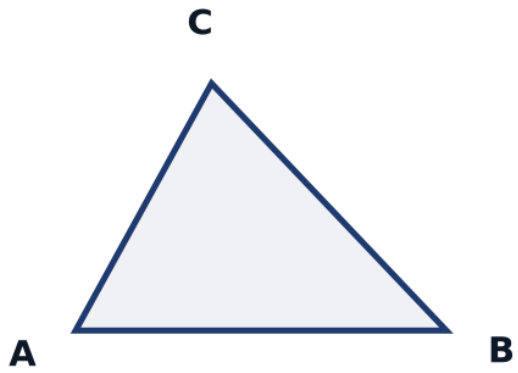
On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(2)$.



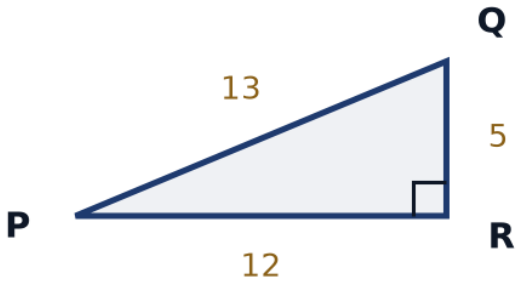
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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for trigonometric identities consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 6x + 9$ in solving trigonometric equations 1, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for solving trigonometric equations 1 consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

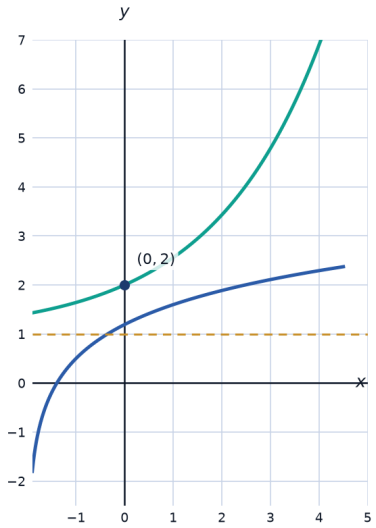
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Calculus (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

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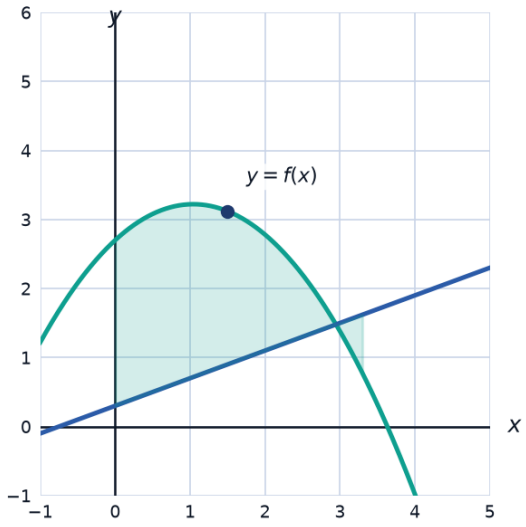
10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

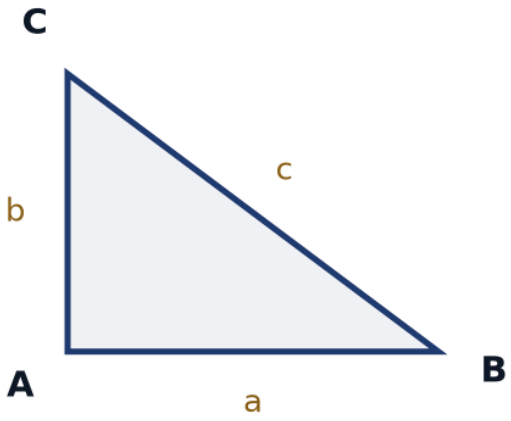
Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



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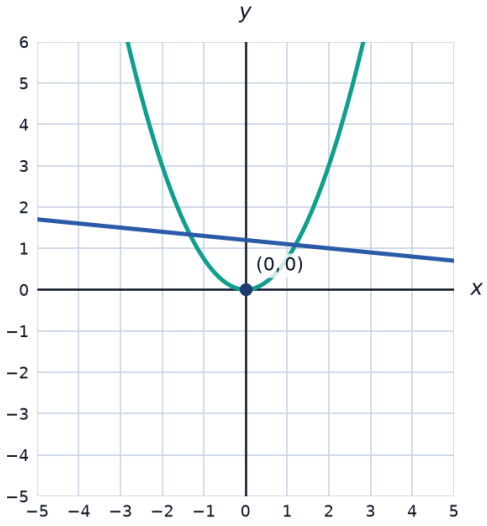
12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

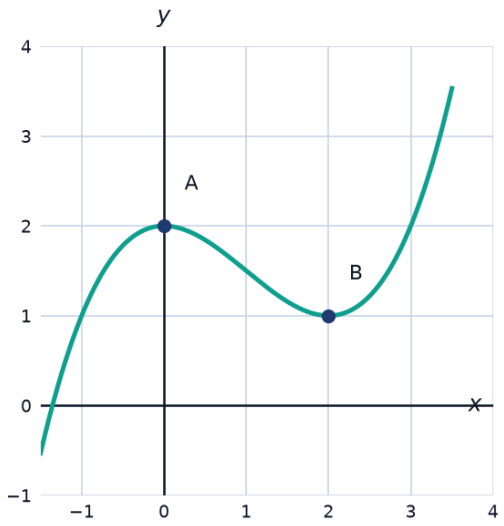
If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

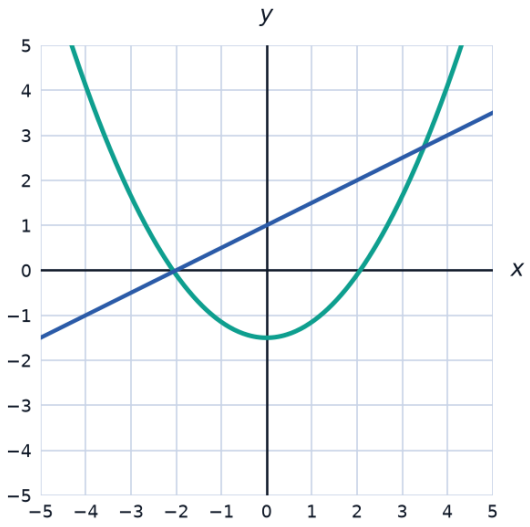
Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

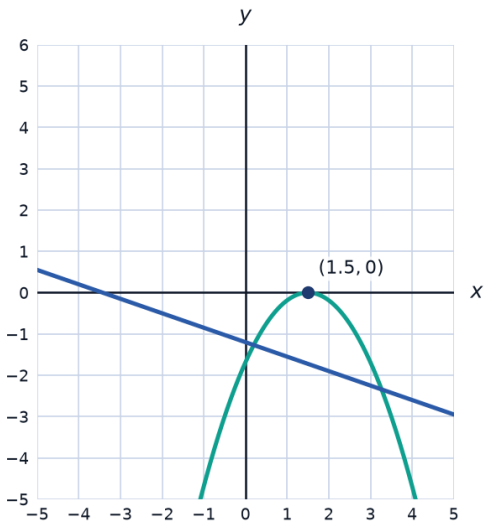
The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.



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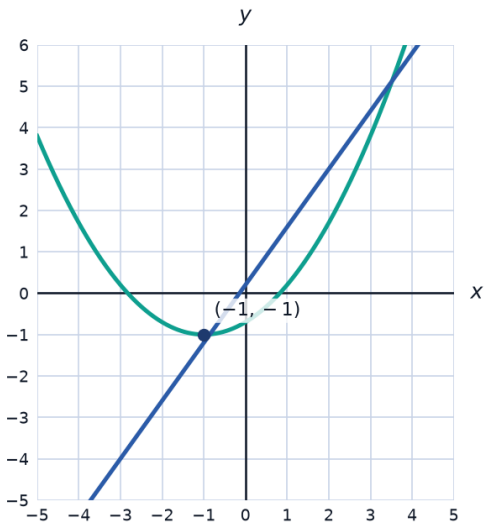
16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 AI SL — Specific Topics

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number & Algebra

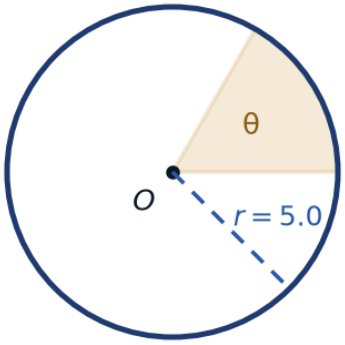
Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Geometry & Trigonometry

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Statistics & Probability

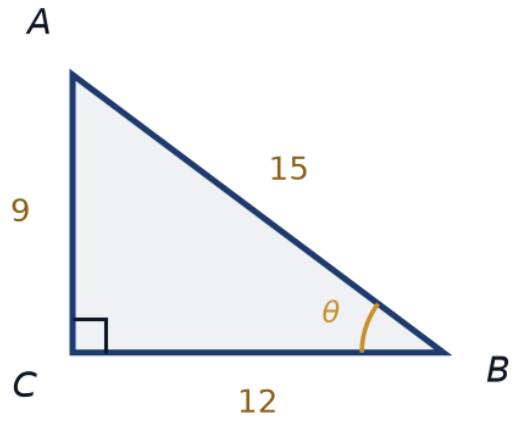
Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying exact values in right-angled triangles, $m\angle A = 35^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 100°

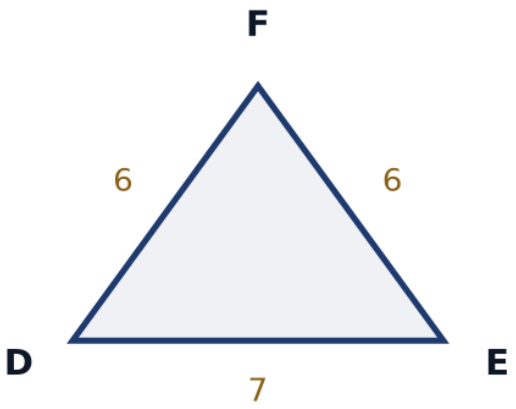
b. 110°

c. 92°

d. 145°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(2)$.



a. 12

b. 15

c. 6

d. 16

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for sine rule and the ambiguous case consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

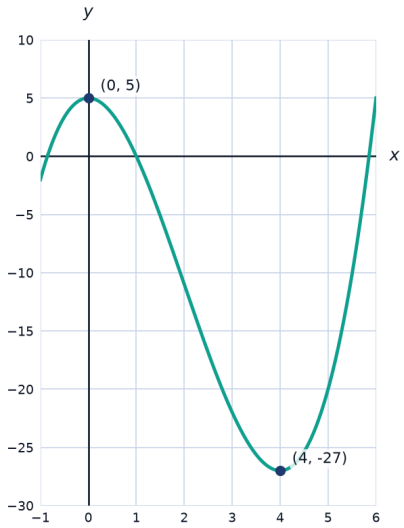
c. 20.5

d. 22.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the function $f(x) = x^3 - 6x^2 + 5$. The graph of $y = f(x)$ is shown below. Determine the coordinates of the local maximum of $f(x)$.



a. (0, 5)

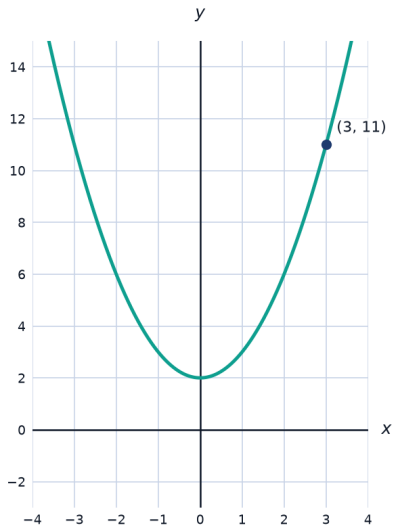
b. (4, -27)

c. (2, -11)

d. (1, 0)

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = x^2 + k$ is shown below. The graph passes through the point $(3, 11)$. Determine the value of k.



a. 2

b. -2

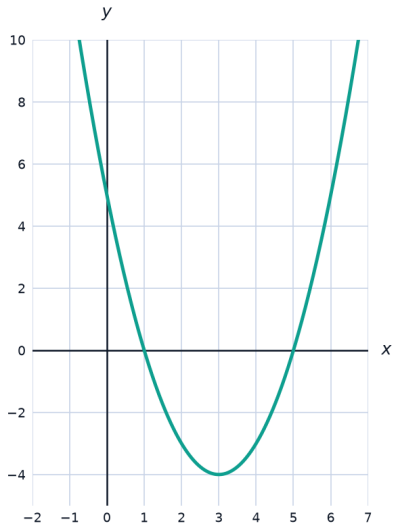
c. 5

d. -5

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the function $f(x) = x^2 - 6x + 5$. Find the x-coordinate of the vertex of the parabola represented by $y = f(x)$.



a. 1

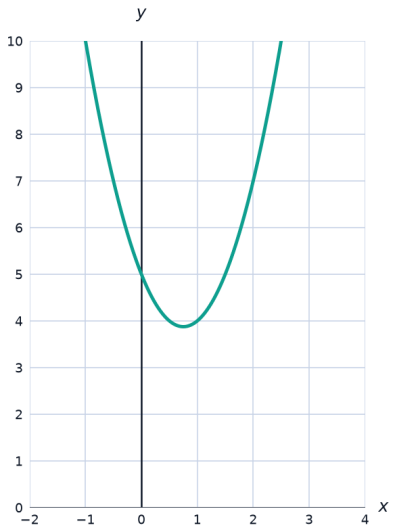
b. 3

c. -1

d. -3

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the function $f(x) = 2x^2 - 3x + 5$ is shown below. Determine the value of $f'(2)$.



a. 7

b. 5

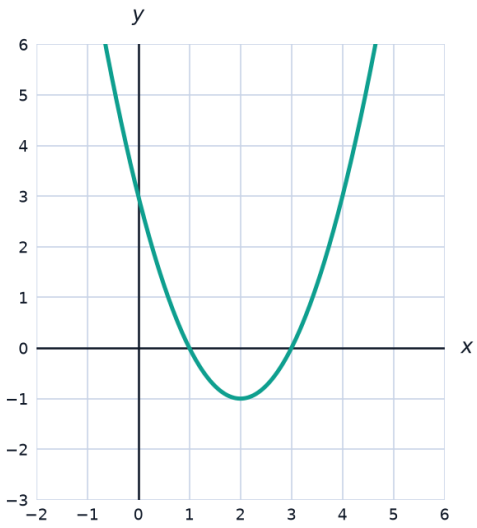
c. 9

d. 2

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. Find the y-intercept of the graph.



a. 3

b. -1

c. 0

d. -3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^3 - 6x^2 + 5$. Find the values of x for which the tangent to the curve $y = f(x)$ is parallel to the x-axis.

a. $x = 0$ or $x = 4$

b. $x = 1$ or $x = 5$

c. $x = 0$ or $x = 6$

d. $x = -1$ or $x = -5$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the tangent to $y = -2x^3 + 2x$ at $x = 1$, in the form $y = mx + c$.

a. $y = -4x + 6$

b. $y = -4x$

c. $y = 4x + 4$

d. $y = -4x + 4$

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the x-coordinates of the stationary points of $f(x)$, given $f'(x) = 2x^2 - 8x + 6$. Find the larger x-coordinate.

a. 2

b. 0

c. 3

d. 1

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find the area enclosed by the curve $y = x^2$, the x-axis, and the line $x = 3$, for $0 \le x \le 3$.

a. 9

b. 5

c. 4

d. 6

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the gradient of the tangent to $y = -4x^3 - 5x^2 - 5x + 8$ at $x = 0$.

a. 0

b. -15

c. 5

d. -5

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Tangent to $y = x^3$ at $(2, 8)$: equation:

a. $y = 12x - 16$

b. $y = 12x + 8$

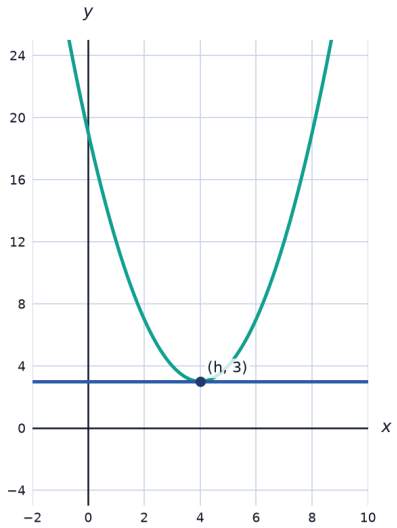
c. $y = 6x - 4$

d. $y = 8x - 8$

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35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The graph of the quadratic function $f(x) = x^2 - 8x + k$ has its vertex at the point $(h, 3)$. Determine the value of k.



a. 13

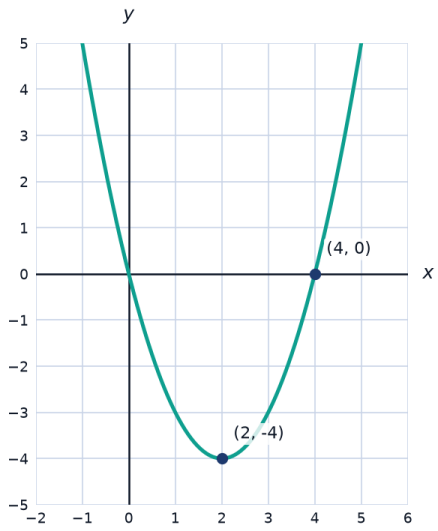
b. 19

c. 22

d. 25

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at (2, -4) and it passes through the point (4, 0). Determine the value of c.



a. -4

b. 0

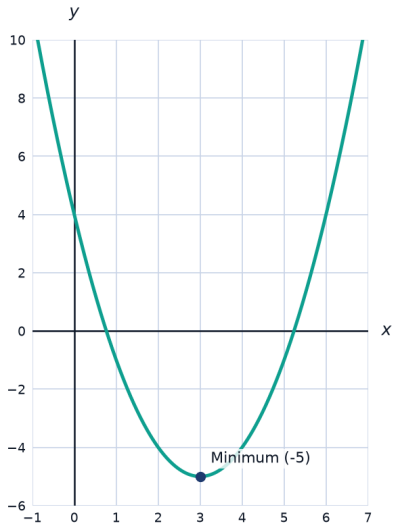
c. 2

d. 4

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37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown below. The graph has a minimum value of -5. Determine the value of k.



a. 4

b. 5

c. 14

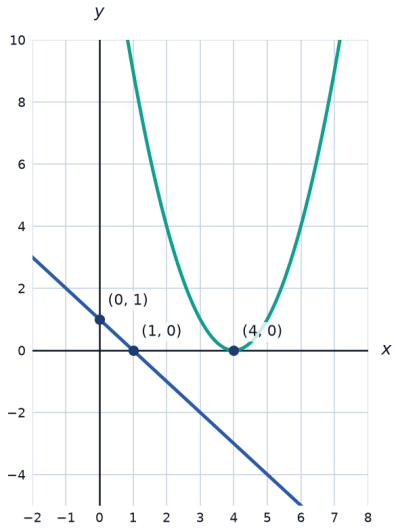
d. 3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions

The graph of the quadratic function $f(x) = x^2 - 8x + k$ is tangent to the x-axis. The graph of the linear function $g(x) = -x + 1$ is also shown. Find the value of k.



a. 16

b. 8

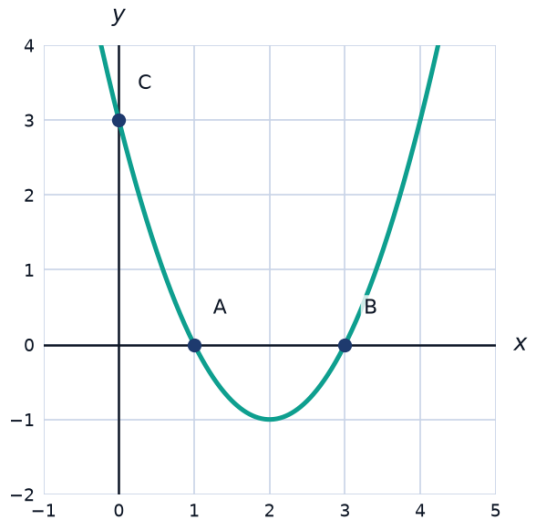
c. 4

d. 2

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39 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 3.01 Geometry & Trigonometry**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the sum of the x-coordinates of A and B.



a. -4

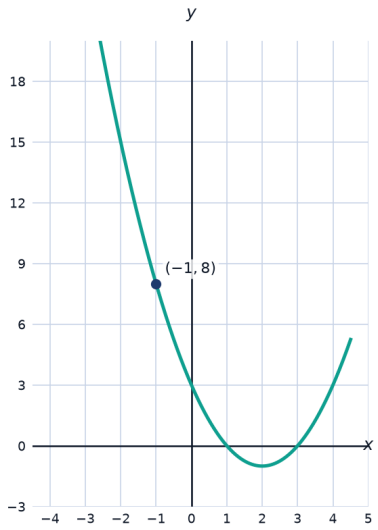
b. 3

c. 4

d. 1

40 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.01 Statistics & Probability**

Consider the function $f(x) = x^2 - 4x + 3$. Find the sum of the x-intercepts of the graph of $y = f(x)$.



a. 4

b. 3

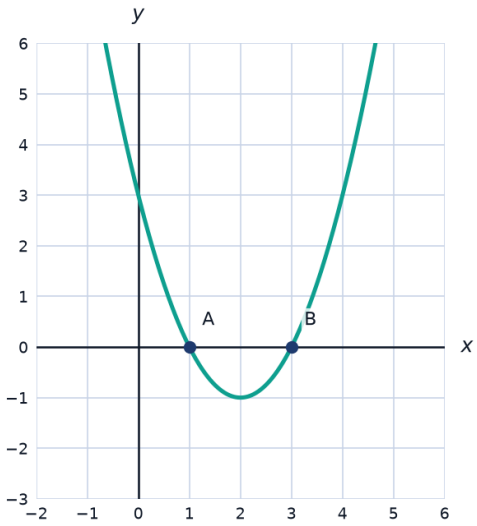
c. -4

d. 1

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



a. 1

b. 2

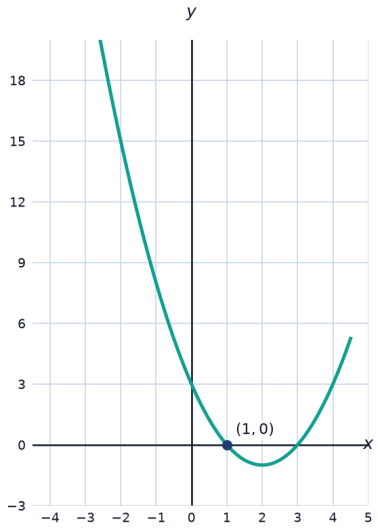
c. 3

d. 4

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42 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 1.01 Number & Algebra**

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(5)$.



a. 8

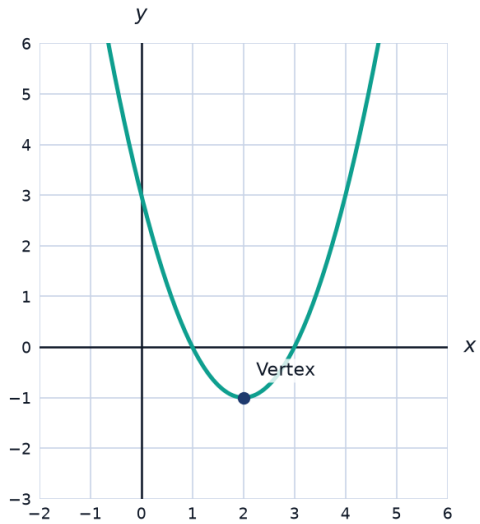
b. 1

c. 0

d. 18

43 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.01 Functions**

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.



a. 2

b. -2

c. 1

d. 3

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Geometry & Trigonometry

A particle moves in the xy-plane. Its velocity vector $\mathbf{v}(t)$ at time t is given by $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$, where $t \geq 0$. The initial position of the particle at $t = 0$ is $\mathbf{r}(0) = \langle 5, -2 \rangle$. Find the position vector $\mathbf{r}(t)$ of the particle at time t. Then, determine the value of t for which the particle's speed is minimized and state the minimum speed.

a. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{2}$ at $t = 1$

b. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{5}$ at $t = 0$

c. $\mathbf{r}(t) = \langle 3t^2 - 6t + 5, e^t - t - 2 \rangle$, minimum speed 0 at $t = 0$

d. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{3}$ at $t = 0$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Statistics & Probability

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

a. $y = 2x - 5$

b. $y = -\frac{1}{2}x + 5$

c. $y = 2x - 8$

d. $y = -2x + 11$

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